

Scalability & Future Plan

Date	1 July 2026
Team ID	SWTID-2026-4523
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address
1	The application is currently deployed only on a local Flask server.	Users cannot access the application remotely.	High
2	The prediction model is trained on a limited dataset.	Prediction accuracy may decrease for unseen or diverse data.	Medium
3	The system does not support multiple users simultaneously.	Performance may reduce with high user traffic.	Medium

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a single/local user.	Deploy on a cloud platform (Render, Railway, or AWS) to support multiple users.

2	Data Storage	Uses a static dataset without a database.	Integrate a MySQL or PostgreSQL database to manage applicant records.
3	Performance	Suitable for small-scale usage.	Optimize the model and use a production server such as Gunicorn for better performance.
4	Security	Basic input validation is implemented.	Add user authentication, HTTPS, and secure data handling mechanisms.

Future Roadmap:

Phase	Scalability Aspect	Current State	Proposed Upgrade/ Solution
Phase 1	Deploy the application to a cloud platform.	1–2 Months	Enable online access from anywhere.
Phase 2	Improve the machine learning model using a larger and more diverse dataset.	2–3 Months	Increase prediction accuracy and reliability.
Phase 3	Add user authentication and applicant history management.	3–4 Months	Improve security and provide personalized access.
Phase 4	Develop a mobile-friendly version and REST API integration.	4–6 Months	Improve accessibility and support integration with other applications.